RAJALAKSHMI ENGINEERING COLLEGE

Vignesh V 251001300-2025-IT

V2

Started on	Saturday, 3 October 2026, 10:19 PM
State	Finished
Completed on	Saturday, 3 October 2026, 10:19 PM
Time taken	30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findPairWithDifference(int arr[], int n, int k) {
    int i = 0, j = 1;

    // Two-pointer approach on sorted array
    while (i < n && j < n) {
        if (i != j && arr[j] - arr[i] == k) {
            return 1;
        } else if (arr[j] - arr[i] < k) {
            j++;
        } else {
            i++;
        }
    }

    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) return 0;

    printf("%d\n", findPairWithDifference(arr, n, k));

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vignesh V 251001300-2025-IT

V2

Started on	Saturday, 3 October 2026, 10:18 PM
State	Finished
Completed on	Saturday, 3 October 2026, 10:19 PM
Time taken	30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findPairWithDifference(int arr[], int n, int k) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i != j && arr[j] - arr[i] == k) {
                return 1;
            }
        }
    }
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) return 0;

    printf("%d\n", findPairWithDifference(arr, n, k));

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vignesh V 251001300-2025-IT

V2

Started on	Saturday, 3 October 2026, 10:17 PM
State	Finished
Completed on	Saturday, 3 October 2026, 10:18 PM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1, n2;

    // Read first array
    if (scanf("%d", &n1) != 1) return;
    int a[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &a[i]);
    }

    // Read second array
    if (scanf("%d", &n2) != 1) return;
    int b[n2];
    for (int j = 0; j < n2; j++) {
        scanf("%d", &b[j]);
    }

    int i = 0, j = 0;
    int first = 1;
```

```
    // Two-pointer linear traversal
    while (i < n1 && j < n2) {
        if (a[i] < b[j]) {
            i++;
        } else if (b[j] < a[i]) {
            j++;
        } else {
            // Both elements are equal
            if (!first) {
                printf(" ");
            }
            printf("%d", a[i]);
            first = 0;
            i++;
            j++;
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) == 1) {
        while (t--) {
            solve();
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vignesh V 251001300-2025-IT

V2

Started on	Saturday, 3 October 2026, 10:17 PM
State	Finished
Completed on	Saturday, 3 October 2026, 10:17 PM
Time taken	35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1, n2;

    // Read first array length and elements
    if (scanf("%d", &n1) != 1) return;
    int a[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &a[i]);
    }

    // Read second array length and elements
    if (scanf("%d", &n2) != 1) return;
    int b[n2];
    for (int j = 0; j < n2; j++) {
        scanf("%d", &b[j]);
    }

    int first = 1;

    // O(m * n) check using two nested loops
```

```
    // O(m * n) - Check using two nested loops
    for (int i = 0; i < n1; i++) {
        for (int j = 0; j < n2; j++) {
            if (a[i] == b[j]) {
                if (!first) {
                    printf(" ");
                }
                printf("%d", a[i]);
                first = 0;
                break; // Move to next element in array a
            }
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) == 1) {
        while (t--) {
            solve();
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vignesh V 251001300-2025-IT

V2

Started on	Saturday, 3 October 2026, 10:16 PM
State	Finished
Completed on	Saturday, 3 October 2026, 10:16 PM
Time taken	32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findDuplicate(int arr[], int n) {
    if (n <= 1) return -1;

    // Phase 1: Finding the intersection point in the cycle
    int tortoise = arr[0];
    int hare = arr[0];

    do {
        tortoise = arr[tortoise];
        hare = arr[arr[hare]];
    } while (tortoise != hare);

    // Phase 2: Finding the entrance to the cycle (duplicate value)
    tortoise = arr[0];
    while (tortoise != hare) {
        tortoise = arr[tortoise];
        hare = arr[hare];
    }

    return hare;
}

int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        int arr[n];
        for (int i = 0; i < n; i++) {
            scanf("%d", &arr[i]);
        }
        printf("%d\n", findDuplicate(arr, n));
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vignesh V 251001300-2025-IT

V2

Started on	Saturday, 3 October 2026, 10:15 PM
State	Finished
Completed on	Saturday, 3 October 2026, 10:16 PM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findDuplicate(int arr[], int n) {
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                return arr[i];
            }
        }
    }
    return -1;
}

int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        int arr[n];
        for (int i = 0; i < n; i++) {
            scanf("%d", &arr[i]);
        }
        printf("%d\n", findDuplicate(arr, n));
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course